

MTH 446: Lecture # 7

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Lecture Span

1. Topology of the complex plane

Definition 0.1. A polygonal line is formed by a finite number of line intervals, joined end to end

Definition 0.2. A non-empty set A is called (polygonally) connected if every two points of A can be connected by a polygonal line that lies completely in A . In otherwords, you can draw a continuous line from z_0 to z_1 for every $z_0, z_1 \in A$.

Definition 0.3. A non-empty open set is called domain if it is connected.

Definition 0.4. A domain together with some, none, or all of its boundary points is called a region.

Definition 0.5. Let $A \subseteq \mathbb{C}$ and $z_0 \in \mathbb{C}$. The point z_0 is called an accumulation point of A if

$$\forall \epsilon > 0, \quad \overset{\circ}{D}_\epsilon(z_0) \cap A \neq \emptyset \quad (1)$$

Note, $\overset{\circ}{D}$ is the deleted ϵ neighborhood!!! Therefore, every z_0 is an accumulation point if every deleted ϵ neighborhood of z_0 contains at least one point of A .

A single point is not an accumulation point!

Example

Let $A = \{i\}$. Then prove that $i \in \partial A$. and i is not an accumulation point.

1. We first prove that $i \in \partial A$. In particular, $D_\epsilon(i) \cap A \neq \emptyset$ and $D_\epsilon(i) \cap \mathbb{C}/A \neq \emptyset$. Therefore, $i \in \partial A$.
2. i is not an accumulation point, because $\overset{\circ}{D}_\epsilon(i) \cap A = \emptyset$.

Example

Suppose

$$A = \{z \in \mathbb{C}/\{0\} : 0 \leq \arg(z) < \pi/2\} \quad (2)$$

Then, prove that $z = 0$ is an accumulation point.

Proof. Draw a deleted ϵ neighborhood at 0. From the sketch, we see that $\overset{\circ}{D}_\epsilon(0) \cap A \neq \emptyset$. Therefore, 0 is an accumulation point of A . $\square$

Example

Let

$$A = \{z_n = i^n : n = 1, 2, 3, \dots\} \quad (3)$$

Find the set of all accumulation points.

Proof. It is obvious that $A = \{1, -1, i, -i\}$. Then let $\epsilon_1 = \min_i \{|z_i - i|\}$, then for $\epsilon < \epsilon_1$, we have that $\overset{\circ}{D}_\epsilon(z_0) \cap A = \emptyset$. Therefore, i is not an accumulation point. By symmetry, it follows that all other points of A are not accumulation points. Next, let $z_0 \notin A$. Then, let $\epsilon_2 = \min_i \{|z_i - z_0|\}$, then choose $\epsilon < \epsilon_2$, then $D_\epsilon(z_0) \cap A = \emptyset$. Therefore, z_0 is not an accumulation point. $\square$

Therefore, the set of all accumulation points of is $\emptyset$.

Definition 0.6. A point $z_0 \in F$ is an isolated point of F if

$$\exists \epsilon > 0 : \overset{\circ}{D}_\epsilon(z_0) \cap F = \emptyset$$

For example, all points of the set $A = \{i, 1, -1, -i\}$.

It follows that every point of F is either an accumulation point of F or it is an isolated point of F .

Functions of complex variables

Let $D \subseteq \mathbb{C}$ be non empty. Then $w = f(z)$, where $z \in D$ and $w \in \mathbb{C}$ is called the function of the complex variable, i.e.

$$f : D \subseteq \mathbb{C} \rightarrow \mathbb{C} \quad (4)$$

D is called the domain of definition.

Example

$$f(z) = \frac{1}{1 + z^2} \quad (5)$$

Then the domain of f is $\mathbb{C}/\{i, -i\}$.

Example

$f(z) = |z|^2$. Then domain of f is $\mathbb{C}$.

Example

Consider

$$P_n(z) = \sum_i a_i x^i \quad (6)$$

With $a_i \in \mathbb{C}$ for all i . Then $P_n(z)$ is called a polynomial of degree n . Then, the domain of P_n is $\mathbb{C}$.

Example

$$f(z) = \arg(1/z) \quad (7)$$

Then, the domain of f is $\mathbb{C}/\{0\}$.

Next, set $u(x, y) = \operatorname{Re}(f(z)) = \operatorname{Re}(f(x + iy))$ and $S(x, y) = \operatorname{Im}(f(z))$. Then

$$f(z) = u(x, y) + iS(x, y) \quad (8)$$

Example

Let $f(z) = z^2$. Then $f(z) = (x + iy)^2 = zx^2 + 2xyi - y^2 = (x^2 - y^2) + 2xyi$. Therefore, $u = x^2 - y^2$ and $s = 2xy$.

Example

Let $f(z) = x^2 + 2y + 5xi$. Write f in terms of z . Let $x = (z + \bar{z})/2$ and $y = (z - \bar{z})/2i$. Then plug these substitutions into the function.

Next time, we will discuss the polar coordinates.